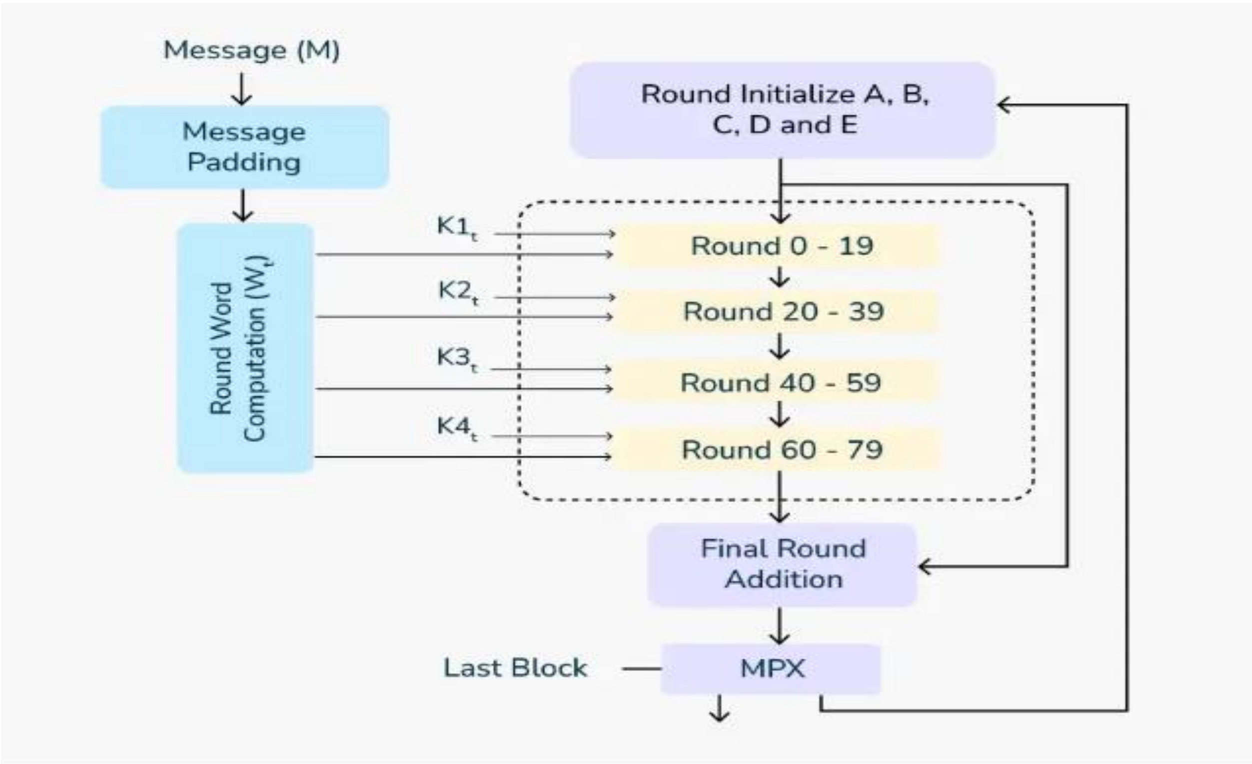
SHA is a family of cryptographic hash functions designed to ensure **data integrity**. It takes an input (message) and produces a fixed-size string of bytes — typically a **digest** that uniquely represents the input.

SHA is **one-way**: you can't reverse the hash to get the original input. Even a tiny change in the input drastically changes the output.

1. **Input Message**: Any length of data.
2. **Padding**: Message is padded to make its length a multiple of 512 bits.
3. **Parsing**: Padded message is split into 512-bit blocks.
4. **Initialization**: Set initial hash values.
5. **Compression Function**: Each block is processed using bitwise operations, modular additions, and logical functions.
6. **Final Output**: A fixed-size hash (e.g., 256 bits for SHA-256).



SHA-1 Hash

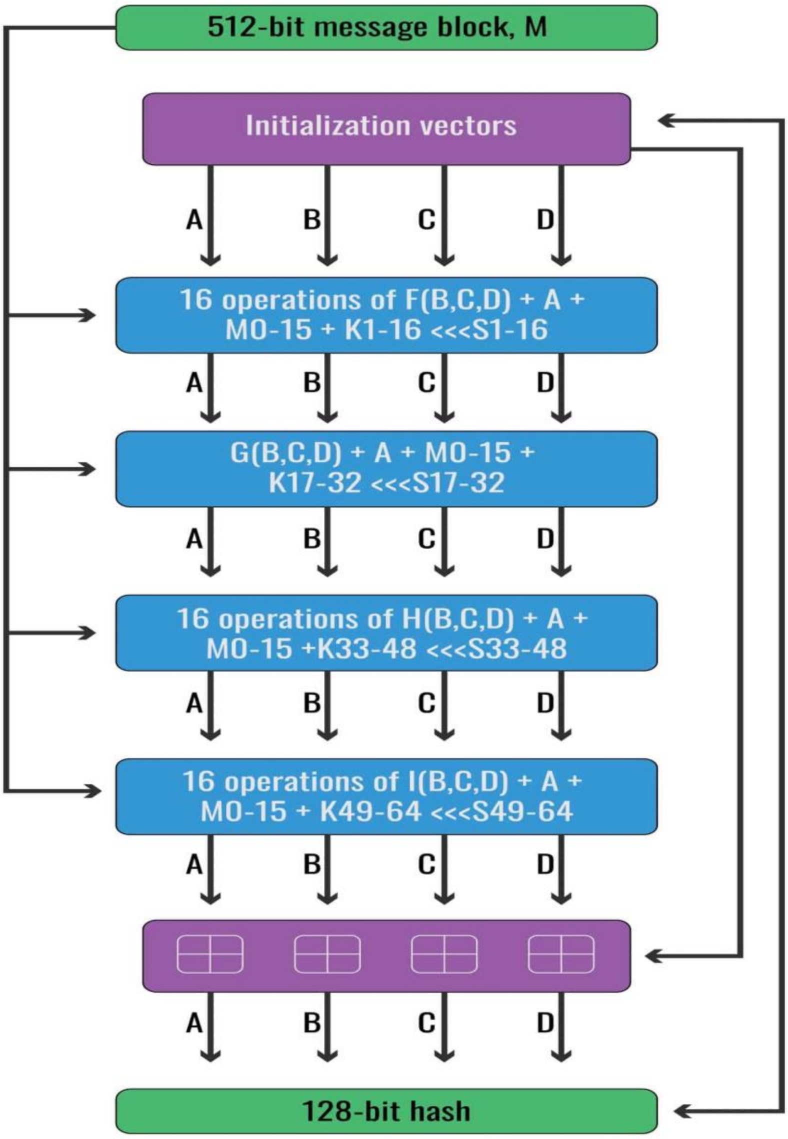
MD5 is a **cryptographic hash function** developed by Ronald Rivest in 1991. It takes an input of any length and produces a **128-bit fixed-length hash** — often used for checksums, data integrity, and fingerprinting files.

Even a tiny change in the input drastically changes the output hash — that's the avalanche effect!

MD5 Algorithm :

1. **Padding the Message**
    - Message is padded so its length is 64 bits less than a multiple of 512.
2. **Appending Length**
    - The original length of the message is added as a 64-bit value.
3. **Initialize MD Buffer**
    - Four 32-bit buffers: A, B, C, D with predefined constants.
4. **Process Message in 512-bit Blocks**
    - Each block goes through **4 rounds** of transformation (16 operations per round).
    - Uses logical functions (AND, OR, XOR, NOT) and modular addition.
5. **Output**

- Final digest is a 128-bit hash (usually shown as 32 hex characters).


512-bit message block, M
Initialization vectors
A
B
C
D
16 operations of F(B,C,D) + A + M0-15 + K1-16 <<<S1-16
G(B,C,D) + A + M0-15 + K17-32 <<<S17-32
16 operations of H(B,C,D) + A + M0-15 +K33-48 <<<S33-48
16 operations of I(B,C,D) + A + M0-15 + K49-64 <<<S49-64
128-bit hash